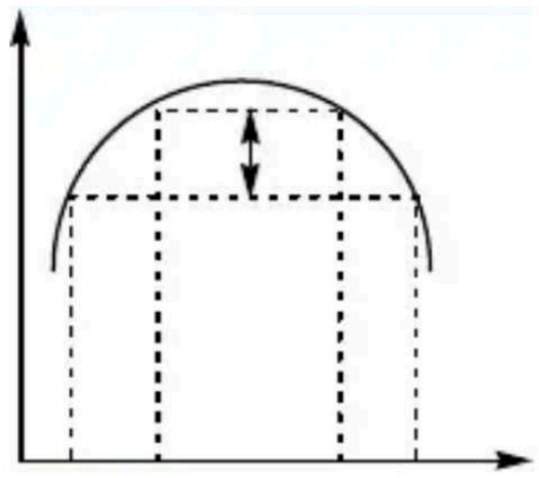
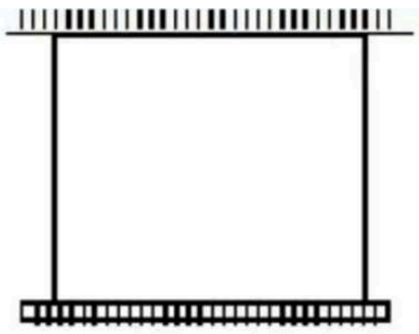
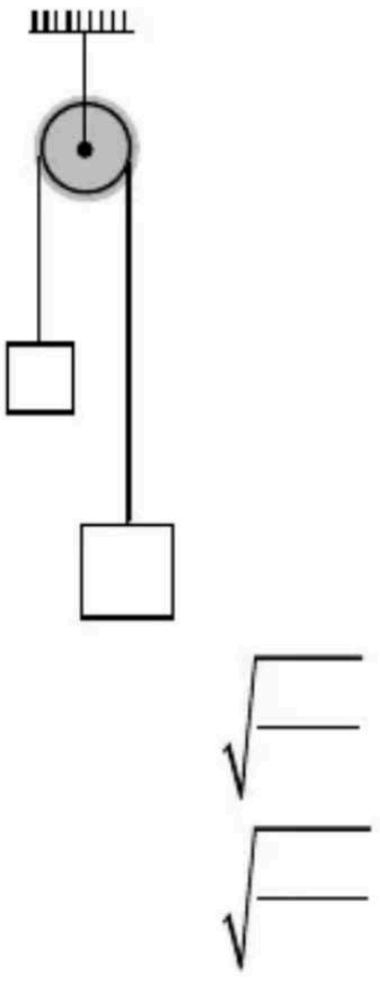
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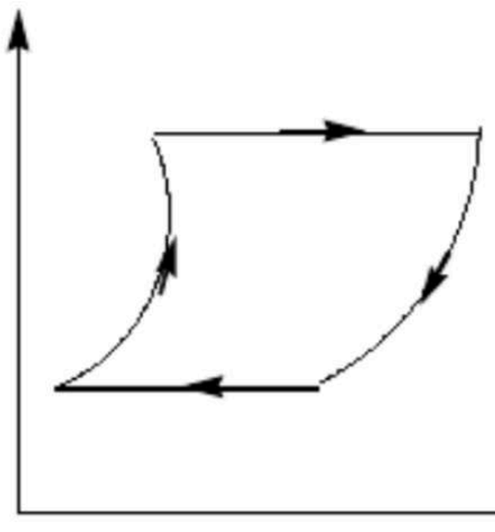
THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

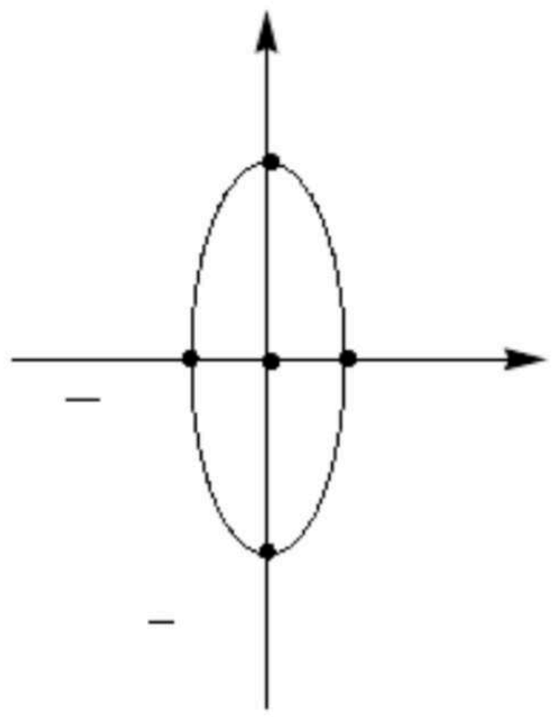
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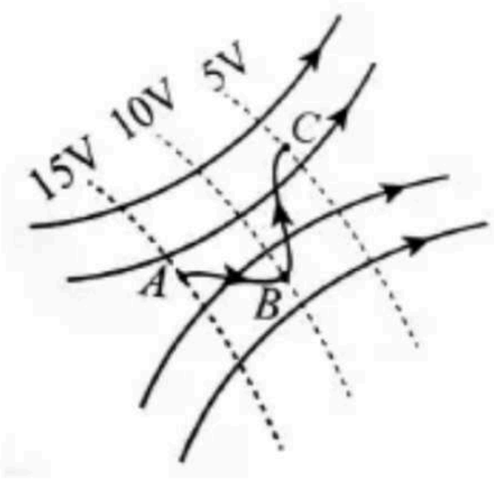


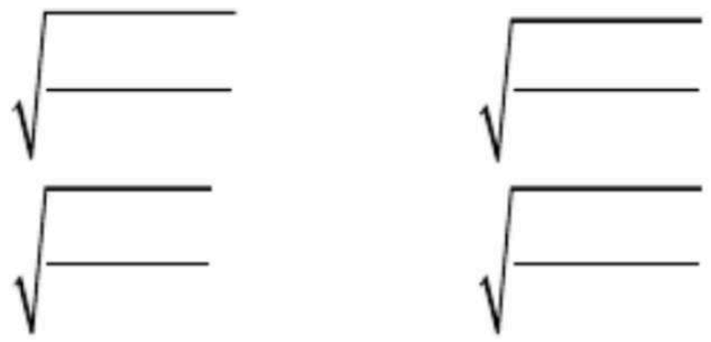


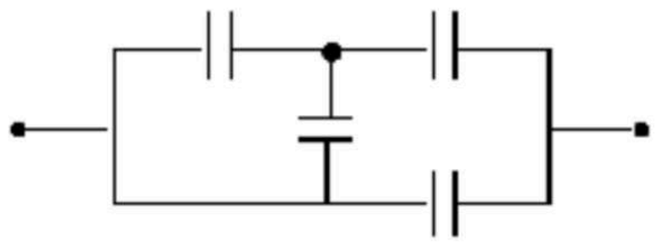




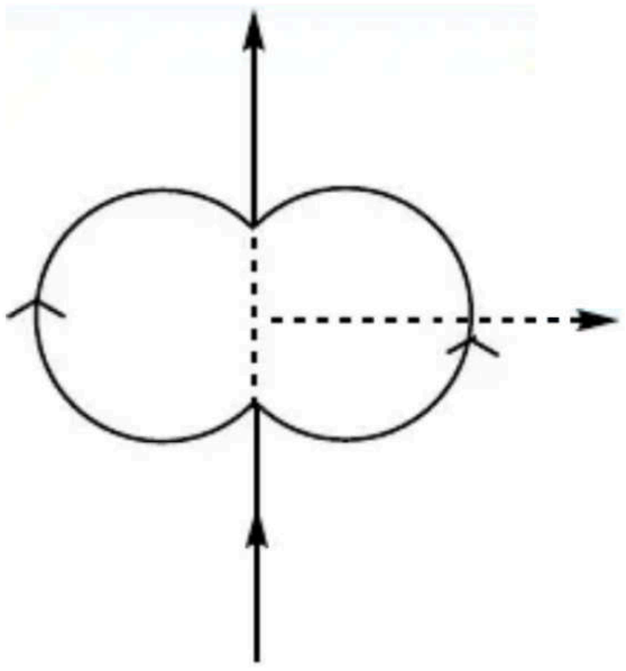


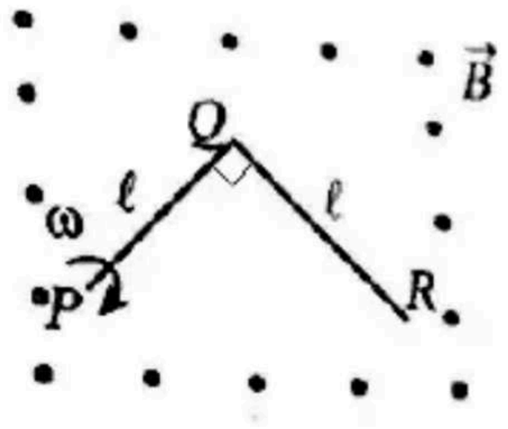

15V
10V
5V
C
A
B

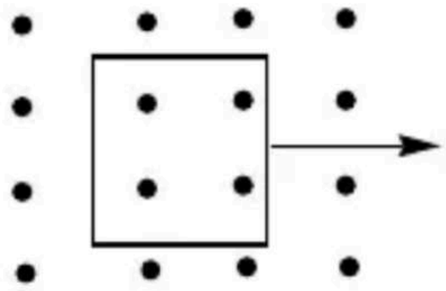




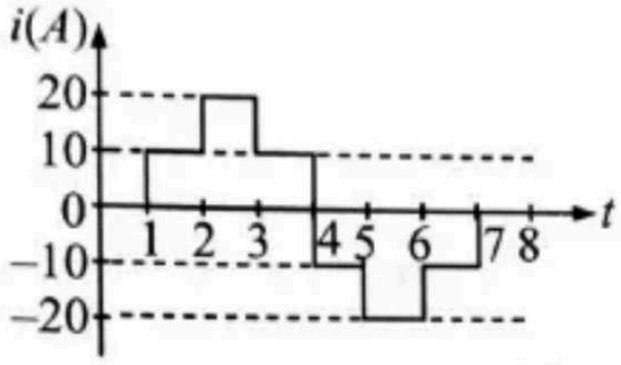





$\vec{B}$
Q
ℓ
ℓ
ω
P
R



31. Figure shows the wave form of an alternating current. Find rms value of the current.



(1) (2) $10\sqrt{2}\ A$

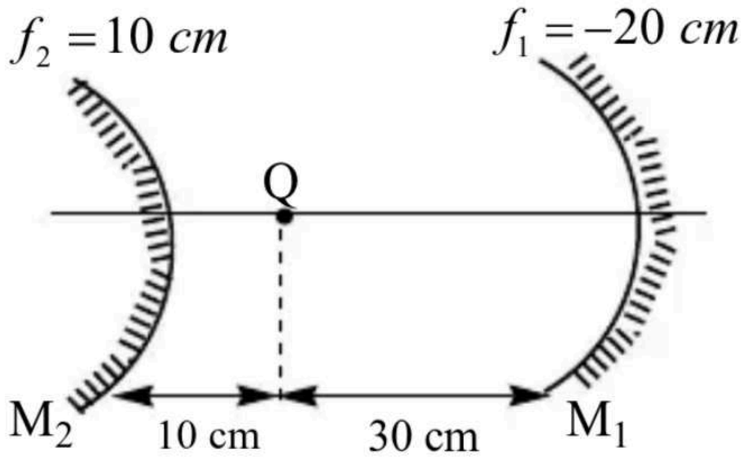
(3) (4) zero

32. An electrical heater and a capacitor are joined in series across a 200 V, 50 Hz AC supply. The potential difference across the heater is $50\sqrt{7}\ V$. The time lag between maximum voltage and maximum current is nearly

(1) 2ms (2) 3ms

(3) 1ms (4) 4ms

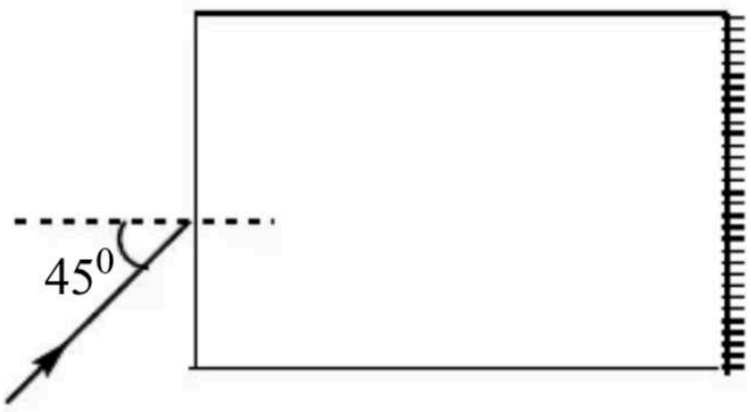
33. In figure there are two mirrors; one convex mirror, f_1 = 20 cm and other of f_2 = 10 cm. Find the total magnification after two reflections first on M_1 and then on M_2.



(1) +4 (2)

(3) +2 (4)

34. Right side face of a glass cube of refractive index $= \sqrt{2}$ is silvered as shown. A ray of light is incident on left face of the cube as shown. The deviation of the ray when it comes out of the glass cube is



(1) 0^0 (2) 90^0

(3) 180^0 (4) 270^0

35. An object is placed in front of a thin convex lens of focal length 30 cm and a plane mirror is placed 15 cm behind the lens. If the final image of the object coincides with the object, the distance of the object from the lens is

(1) 15 cm (2) 25 cm

(3) 30 cm (4) 60 cm

36. In YDSE set-up d = 1 mm, nm and D = 1. Find the minimum distance between two points on the screen having 75% intensity of the maximum intensity.

(1) 0.1 mm (2) 0.2 mm

(3) 0.35 mm (4) 0.4 mm

37. What should the width of each slit be, to obtain 10 maxima of the double slit pattern within the central maximum of the single slit pattern if distance between the slits in double slit experiment is 1mm

(1) 5 mm (2) 0.5 mm

(3) 2 mm (4) 0.2 mm

38. In the Bohr model of the hydrogen atom, let R, V and E represent the radius of orbit, speed of election and the total energy of the electron respectively. Which of the following quantities are proportional of the quantum number 'n' ?

(1) VER (2) RE/V

(3) V/ER (4) E/RV

39. An electron is in an excited state in a hydrogen-like atom. It has a total energy of -3.4 eV. The kE of the electron is E and its De-Broglie wavelength is .

(1) $E = 6.8\ eV, \quad \sim 6.6\times10^{-10}\ m$

(2) $E = 3.4\ eV, \quad \sim 6.6\times10^{-10}\ m$

(3) $E = 3.4\ eV, \quad \sim 6.6\times10^{-10}\ m$

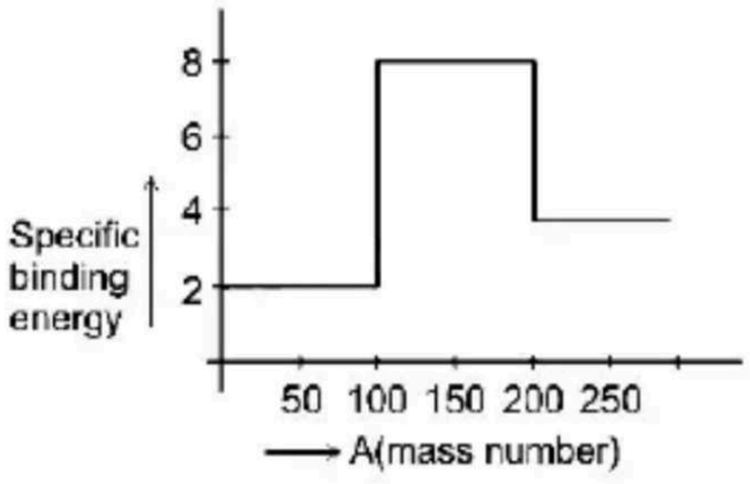
(4) $E = 6.8\ eV, \quad \sim 6.6\times10^{-11}\ m$

40. A metal surface of work function 1.2 eV is irradiated with light of frequency 5.5 x 10^{14} Hz. if the metal surface has a potential of – 2V w.r.t the collector The maximum kinetic energy of electrons on reaching the collector is nearly

(1)1 eV (2) 3 eV

(3) zero (4) 2 eV

41. Assume the graph of specific binding energy versus mass number is as shown in the figure. Using this graph select the correct choice from the following



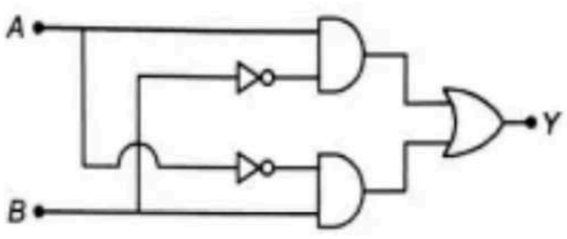
(1) Fission of the nucleus of mass number lying in the range of $100 < A < 200$ will release energy when broken into two fragments

(2) Fusion of two nuclei of mass number laying in the range of $1 < A < 50$ will release energy

(3) Fusion of two nuclei of mass number lying in the range of $51 < A < 100$ will release energy

(4) Fusion of two nuclei of mass number lying in the range of $100 < A < 200$ will release energy

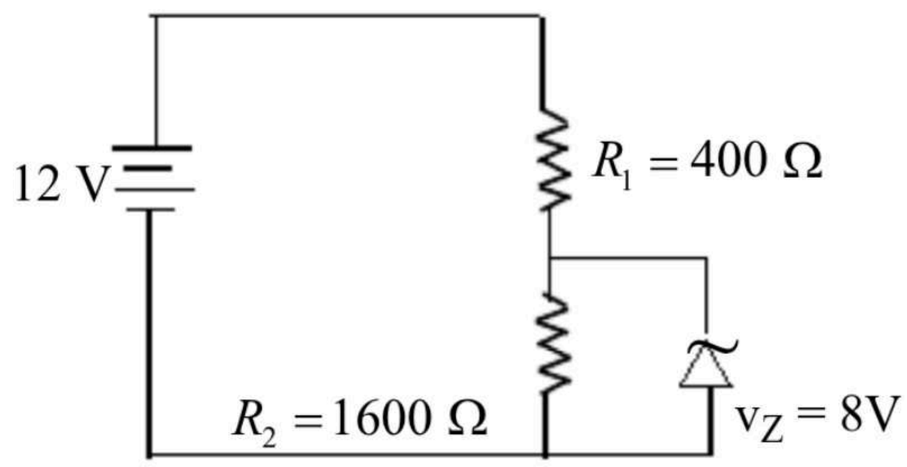
42. For the logic circuit given below, the output Y for A = 0, B = 0 and A = 1, B = 1 are



(1) 0 and 1 (2) 0 and 0

(3) 1 and 0 (4)1 and 1

43. In the circuit shown in figure. Zener diode is properly biased. Find power dissipated in diode.



(1) 40 mW (2) 50 mW

(3) 60 mW (4) 80 mW

44.

	set-I (waves)		set-II (Production)
A.	Infra-red	P	Rapid vibration of electrons in aerials
B.	Radio wave	Q	Electrons in atoms emit photons when they move from higher to lower energy level
C.	Light	R	Klystron valve
D.	Microwave	S	Vibration of atoms and molecules

(1) A-P, B-R, C-S, D-Q

(2) A-S, B-P,C-Q,D-R

(3) A-Q, B-P,C-S,D-R

(4) A-S, B-R,C-P,D-Q

45. A parallel plane capacitor consists of two circular plates of radius R = 0.1m. They are separated by a distance d = 0.5 mm. If electric field between the capacitor plates changes as $\frac{dE}{dt} = 5\times10^3 \frac{V}{m\times s}$ find displacement current between the plates

(1) 10.9 A (2) 11.9 A

(3) 12.9 A (4) 13.9 A

CHEMISTRY

46. Number of structurally isomeric carboxylic acids and esters possible with the molecular formula, $C_4H_8O_2$ is

(1) 4 (2) 5

(3) 7 (4) 6

47. Number of lone pairs in XeF_2 is 'P'. Number of angular nodes for 5f orbital is 'Q'. Sum of 'P' and 'Q' is

(1) 6 (2) 12

(3) 4 (4) 10

48. Solubility product of Zirconium phosphate is 6912 s^n. Value of 'n' is

(1) 6 (2) 8

(3) 7 (4) 5

49. Identify correct statements.

A) Hydroxides of Lanthanoids are definite compounds but not hydrated oxides.

B) $Sm^{2+}(aq)$ acts like a reducing agent

C) TbO_2 acts like an oxidizing agent

D) Compounds of actinoids in "+3" and "+4" oxidation states tends to hydrolyse

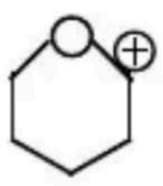
(1) A, B, C and D (2) Only C and D

(3) A and C only (4) B and C only

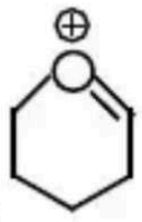
50. Identify correct statements

A) rearrangement of free radicals is a characteristic property in free radical reactions

B)



is more stable than



C) Stability of

$$C_6H_5\overset{\ominus}{C}HCH_3 < C_6H_5\overset{\ominus}{C}H_2 < CH_3\overset{\ominus}{C}H_2 < \overset{\ominus}{C}H_3$$

D) Benzyne is aromatic

(1) A,B and C only (2) B and D only

(3) C and D only (4) A,C and D only

51. CFSE of $[NiCl_4]^{2-}$ is

(1) $-0.8\ \Delta_t$ (2) $-1.2\ \Delta_t$

(3) zero (4) $-2.8\ \Delta_t$

52. 125 grams of 80% pure $CaCO_3$ on thermal decomposition produces 'X' grams of 70% pure CaO. Value of 'X' is

(1) 61.25 g (2) 109.375 g

(3) 39.2 g (4) 80 g

53. $SiCl_4(X)\xrightarrow[-4HCl]{+4H_2O} Y \xrightarrow{1000^0} Z.$

Hybridisation of 'Si' in 'X', 'Y' and 'Z' is respectively.

(1) sp^3, sp, sp (2) sp^3d, sp^3, sp^3

(3) sp^3,sp^3,sp^3 (4) sp^3,sp^3,sp

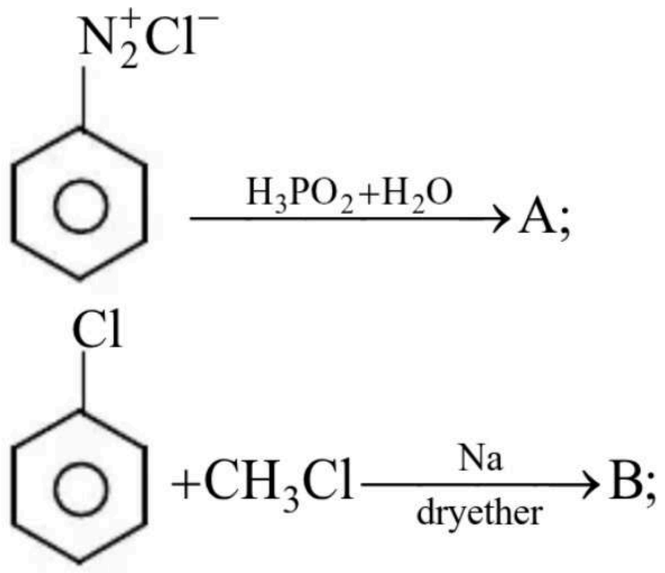
54.

Column – I (Reaction	Column-II (Product)
I) Isobutane $\xrightarrow{KMnO_4}$	A) Methanol
II) $CH_4 \xrightarrow[Mo_2O_3,\Delta]{O_2}$	B) 2-Methylpentane
III) n – hexane $\xrightarrow[an.AlCl_3]{HCl}$	C) Methanal
IV) 2^0butyl chloride $\xrightarrow[dryether]{Na}$	D)3^0 butyl alcohol
	E) 2,5-Dimethylhexane
	F) Isobutyl alcohol
	G) 3-Methylpentane
	H) 3,4-Dimethylhexane

Correct match is

	I	II	III	IV
(1)	F	A	B	H
(2)	D	C	G	H
(3)	D	A	B	E
(4)	F	C	G	E

55.



Identify incorrect statement.

(1) Mixing of liquids 'A' and 'B' does not involve any enthalpy changes

(2) Mixing of liquids 'A' and 'B' leads to increase in randomness

(3) P_A^0 (y-axis) vs X_A (x – axis) is a straight line with positive y-intercept (P_A^0 = VP of A and X_A = mole fraction of A in liquid mixture)

(4) A-B interactions are Hydrogen bonds.

56. Regarding the element with Z = 56 the incorrect statement is
(1) Forms sparingly soluble sulphate
(2) It's chloride is used in the confirmation of carbonate ion
(3) Imparts Crimson red colour in flame test
(4) It's chromate is a yellow coloured sparingly soluble salt

57. Molarity of pure water is
(1) $\frac{1000}{18}$ M (2) 5.55 M
(3) 555.5 M (4) 1 M

58. Identify incorrect statements from the following
A) Glycerol is purified by fractional distillation
B) Column chromatography is an example of descending chromatography
C) In Leibig's method of estimating carbon present in organic compound, CO_2 is absorbed using lime water
D) Sodium nitroprusside is $Na_4\left[Fe(CN)_5NOS\right]$
(1) A,C and D only (2) B and D only
(3) A and B only (4) B,C and D only

59. Number of oxygen atoms per mole of Glutamine is
(1) N_A (2) $2N_A$
(3) $3N_A$ (4) $4N_A$

60. Assertion (A) : Fluorine is the strongest oxidizing agent in aqueous solution (among commonly occurring substances)
Reason (R) : Fluorine is the most electronegative element in the periodic table.
(1) A and R are true, R does not explain A
(2) A and R are true, R explains A
(3) A is true but R is false
(4) A is false but R is true

61. Assertion (A) : Propene reacts with HCl in presence of Benzoyl peroxide to give secondary propyl chloride as major product.
Reason (R) : Secondary propyl free radical is more stable than n-propyl free radical
(1) A and R are true, R explains A
(2) A is false but R is true
(3) A is true but R is false
(4) A and R are true, R does not explain A

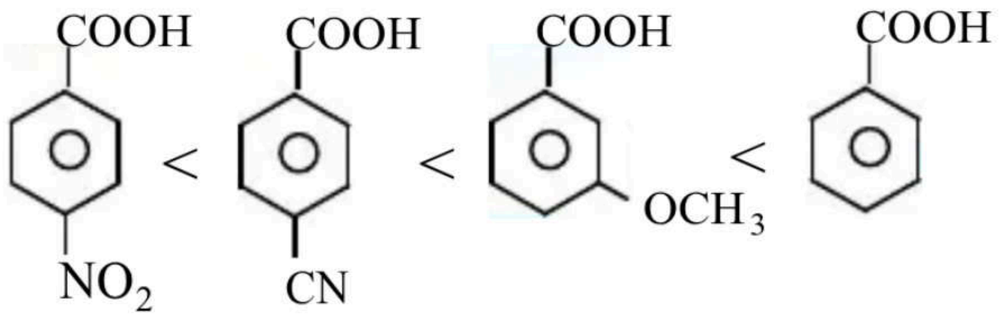
62. Number of fundamental particles in Deutron is 'X'. Number of fundamental particles in α – particle is 'y'. $\frac{x}{y} =$
(1) $\frac{3}{6}$ (2) $\frac{2}{6}$
(3) $\frac{2}{4}$ (4) $\frac{3}{4}$

63. Identify incorrect trend
(1) $CH_3CH_2CH_2CH_2OH > CH_3CH_2COCH_3 > CH_3CH_2CH_2CH_2CH_3 > CH_3CH_2OCH_2CH_3$ .....Boiling point
(2) $CH_3-O-CH_2Br > C_6H_5CH_2Br > (CH_3)_2CHCH_2Br > CH_3CH_2CH_2CH_2Br$ .....Reactivity towards SN^1
(3) $CH_3CH_2CH_2CH_2OH > (CH_3)_2CHCH_2OH > CH_3CH(OH)CH_2CH_3 > (CH_3)_3COH$ .....solubility in water
(4) $CH_3Br > CH_3CH_2Br > (CH_3)_2CHBr > (CH_3)_3CBr$ .....reactivity towards SN^2

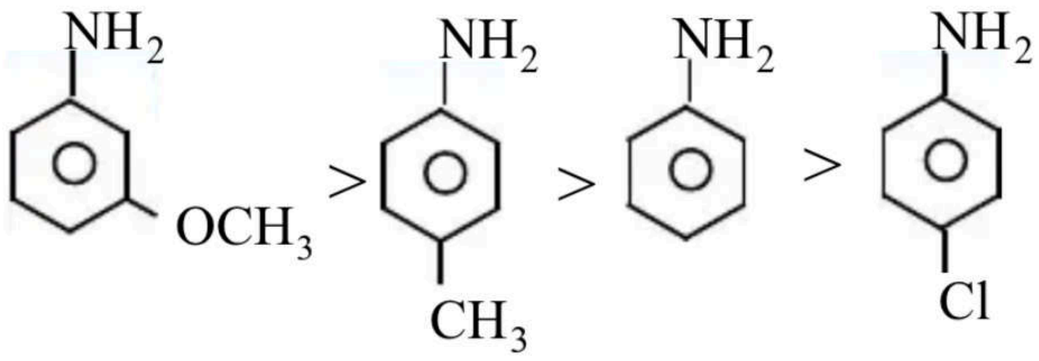
64. Ka_1, and Ka_2 of H_2S are 10^{-6} and 10^{-14} respectively. What is the concentration of bisulphide ion in a mixture containing 0.1M HCl and 0.3 M H_2S (assume ionization of bisulphide ion to be negligible)
(1) 3×10^{-6} M (2) 3.33×10^{-7}
(3) 3×10^{-10} (4) 10^{-5} M

65. Statement-I : In a chemical reaction, equivalent weight of a chemical species can never be greater than its molecular weight.
Statement-II : Excess of thyroxine in body leads to goiter
(1) S-I and S-II are correct
(2) S-I is incorrect but S-II is correct
(3) S-I is correct but S-II is incorrect
(4) S-I and S-II are incorrect

66. Statement-I : p^{k_a} of



Statement-II : Basic strength of



(1) S-I and S-II are correct
(2) S-I is correct but S-II is incorrect
(3) S-I is incorrect but S-II is correct
(4) S-I and S-II are incorrect

67. Electronegativity of Antimony on Pauling scale is 1.9. Electronegativity of Bismuth on the same scale would be
(1) 1.8 (2) 2
(3) 2.1 (4) 1.9

68. $C_2H_5OH, CH_3CHO, CH_3COCH_3,$ $HCOOH, CH_3CHOHCH_3.$

Number of species which give positive Tollen's test is 'X'. Number of species which give yellow precipitate with $Na_2CO_3 + I_2$ is 'Y'. Number species which not only give positive haloform test but also give positive 2,4-DNP test is 'Z'. Incorrect relation is
(1) $XYZ = 18$ (2) $X + Z = Y$
(3) $\frac{Y}{Z} = X$ (4) $X^2 + Y = 4Z$

69. $A(g) \rightarrow 2B(g) + C(g)$ follows first order kinetics. Initial pressure of 'A' is 630 mm. Pressure after 50 minutes is 1260 mm. Incorrect statement is
(1) Half-life of the reaction is 3000 seconds
(2) Time taken for 99.9% completion of the reaction is nearly 500 minutes
(3) Rate constant of the reaction is $0.01386\ s^{-1}$
(4) Initial rate is equal to 8.7318 mm/min

70. Statement-I : RnF_2 is the only compound of Radon that has been isolated
Statement-II : XeF_2 is a planar molecule
(1) S-I and S-II are correct
(2) S-I and S-II are incorrect
(3) S-I is correct but S-II is incorrect
(4) S-I is incorrect but S-II is correct

71. Aniline $\longrightarrow$ Acetanilide. The correct sequence of reagents used are
(1) $NaNO_2 + HCl / 278K;\ HBF_4;\ Cu / NaNO_2, \Delta;$ $Sn / HCl;\ CH_3COCl /$ base
(2) $CH_3COCl / AlCl_3$
(3) $C_6H_5COCl / AlCl_3$
(4) $NaNO_2 + HCl / 278K;\ Cu / NaNO_2,$ $\Delta;\ H_2 / Pt; (CH_3CO)_2O / AlCl_3$

72. Statement-I : Valence bond theory explains paramagnetic behaviour of oxygen
Statement-II : The bonds present in KHF_2 are Ionic, covalent and dative bonds.
(1) S-I is incorrect but S-II is correct
(2) S-I and S-II are correct
(3) S-I is correct but S-II is incorrect
(4) S-I and S-II are incorrect

73. A)

$$HC \equiv CH + NaOH \rightarrow HC \equiv \overset{\ominus\ \oplus}{C}Na + H_2O$$

B)

$$C_2H_5OH + NaOH \rightarrow C_2H_5\overset{\ominus\ \oplus}{O}Na + H_2O$$

C)

$$C_6H_5OH + NaHCO_3 \rightarrow C_6H_5\overset{\ominus\ \oplus}{O}Na + H_2O + CO_2$$

Number of reactions that are feasible are
(1) 3 (2) 0
(3) 1 (4) 2

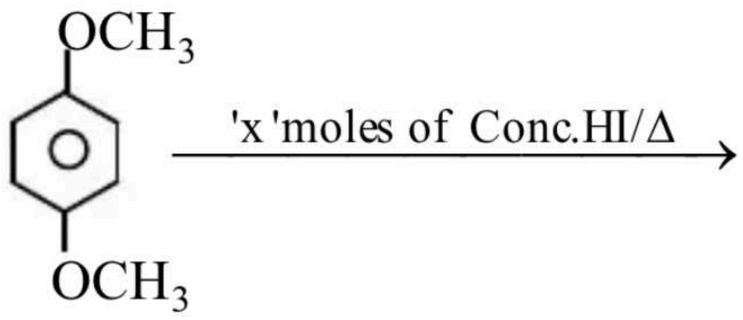
74. Equilibrium constant for a process is 100. The value of change in entropy of the universe under standard conditions will be
(1) 4.606 R (2) 2.303 R (298)
(3) 460.6 R (4) 230.3 R (298)

75. A hydrogen electrode is in contact with a solution containing a mixture of 0.1 M weak mono protic acid, HA and 0.01 M BA (BOH = strong base). If this Hydrogen electrode is connected to a zinc electrode immersed in 0.01 M $ZnSO_4$ solution, the emf of the cell is

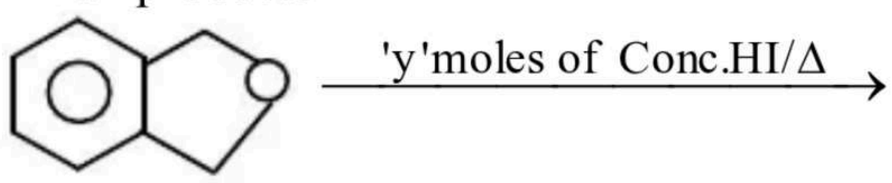
$\left(E^0_{Zn^{2+}} = -0.756\ V \text{ and } p^{k_a} \text{ of } HA = 10^{-5}\right)$

(1) 1.05 V (2) 0.579 V
(3) 0.46 V (4) 1.17 V

76.



Final products



Final products. $\frac{x}{y} =$

(1) 2 (2) 4
(3) 1 (4) 1.5

77. For the reaction, $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ if the degree of dissociation of N_2O_4 is 0.5 and K_P is 2 atm. Equilibrium pressure will be

(1) 1 atm (2) 1.5 atm
(3) 2 atm (4) 2.5 atm

78. $X \xrightarrow[358K]{20\%H_3PO_4}$ Isobutene ' 'X' is

(1) Butan-2-ol
(2) 2-Methyl propanal
(3) 2-Methyl propan-1-ol
(4) 2-Methyl propan-2-ol

79.

Column-I (Complex)	Column-II (Colour)
I) $\left[Ni(H_2O)_6\right]^{2+}$	A) Yellow
II) $\left[Cr(H_2O)_6\right]Cl_3$	B) Red
III) $\left[Co(NH_3)_5 ONO\right]^{2+}$	C) Purple
IV) $\left[Co(NH_3)_4 Cl_2\right]Cl$	D) Green
	E) Pale blue
	F) Violet

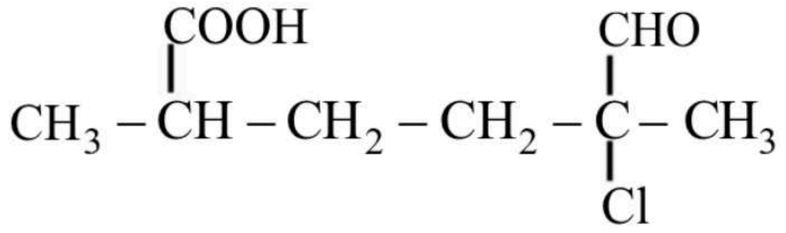
Correct match is

	I	II	III	IV
(1)	F	F	B	F
(2)	D	F	A	D
(3)	D	F	B	D
(4)	F	C	B	F

80. Enthalpy of neutralization of one gram equivalent each of oxalic acid and sodium hydroxide is -54 kJ/mole. Enthalpy of complete ionization of one mole of oxalic acid is

(1) +13.52 kJ (2) +9.99 kJ
(3) +3.33 kJ (4) +6.66 kJ

81. Statement-I : IUPAC name of



is 5-Chloro-6-oxo-2,5-dimethyl hexanoic acid.

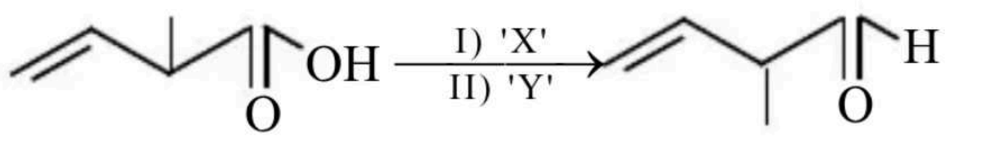
Statement-II : Cinnamaldehyde can exhibit geometrical isomerism

(1) S-I and S-II are correct
(2) S-I is incorrect but S-II is correct
(3) S-I is correct but S-II is incorrect
(4) S-I and S-II are incorrect

82. 'X' is a compound containing Hydrogen and Oxygen. Oxidation state of oxygen in 'X' is -1. 'Y' is the halide with maximum leaving group ability in S_N2 reactions. 'Z' is the strongest base formed by a third period element. Decomposition 'X' in presence of 'Y' as catalyst and 'Z' as medium follows ____order kinetics.

(1) 0 (2) 1
(3) 2 (4) 3

83.



'X' and 'Y' are respectively.

(1) $NaBH_4; CrO_3 / CH_2Cl_2$
(2) $LiAlH_4$;Jone's reagent
(3) H_2 / Pt;PCC
(4) $LiAlH_4$;PCC

84. Equivalent weight of Ferric oxalate acting like reducing agent is (M = Molar mass of Ferric oxalate)

(1) $\frac{M}{4}$ (2) $\frac{M}{1}$

(3) $\frac{M}{6}$ (4) $\frac{M}{2}$

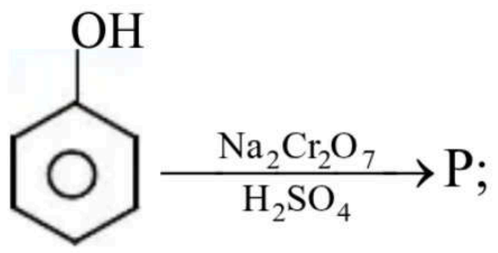
85. $CH_3CH_2COOK(aq) \xrightarrow[\text{electrolysis}]{2F} \underset{\text{(Hydrocarbon)}}{X}$

All the 'X' obtained is subjected to combustion. Δ^0G for this combustion process is

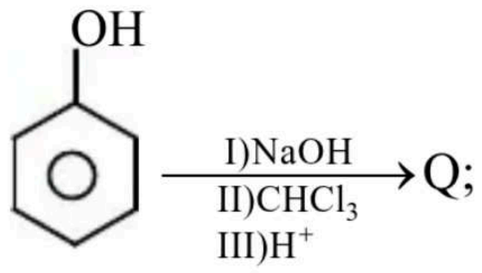
(1) -13 FE^0 (2) -26 FE^0

(3) -4FE^0 (4) -6.5 FE^0

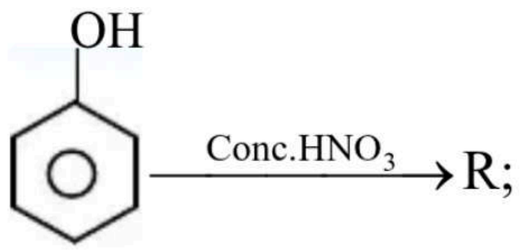
86. A)



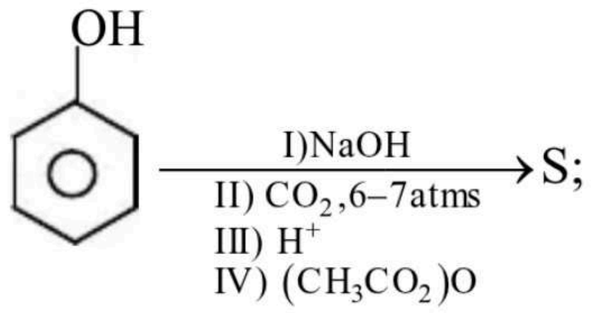
B)



C)



D)



Number of pi bonds are maximum in

(1) P (2) Q

(3) R (4) S

87. The product of uncertainty in velocity and uncertainity in position of a particle is $1.06\times10^{-8} m^2\ s^{-1}$. The mass of the particle $x\times10^{-y}$ kg (nearly). 'x' and 'y' are

(1) 5 and 27 (2) 2 and 26

(3) 5 and 28 (4) 2 and 27

88. $Glucose \xrightarrow[II)HI+RedP/\Delta]{I)Conc.HNO_3} 'X'$. 'X' is

(1) n-hexane (2) Hexanoic acid

(3) n-heptane (4) Glucitol

89. Kernite is $Na_2B_4O_7.xH_2O$. If Argon gas is bubbled through 500 grams of water at 'x' bar then the number of moles of Argon gas dissolved in water will be $(K_H \text{ of } 'Ar' = 40 \text{ K bar})$

(1) $\frac{1}{360}$ (2) $\frac{1}{240}$

(3) $\frac{1}{288}$ (4) $\frac{1}{420}$

90. Titration of $KMnO_4$ vs Mohr's salt is carried out by adding

(1) dil. H_2SO_4 (2) dil. HCl

(3) Conc. HCl (4) Conc. H_2SO_4

BIOLOGY

91. Consider the following

a) A type of specialised connective tissue

b) Present in the outer ear joints

c) Possess solid and pliable matrix

All of the above given features are true for the type of connective tissue which is characterised by the presence of

(1) Osteocytes (2) Lamellae

(3) Chondrocytes (4) Bone marrow

92. One of the following option contains group of organisms included in order Carnivora.

(1) Monkey, Gorilla, Gibbon

(2) Wheat, Rice, Maize

(3) Man, Mango, Monkey

(4) Dog, Tiger, Cat

93. Identify the correct statement about Heterocyst

i) *Nostoc* and *Anabaena* possess them

ii) It can fix atmospheric nitrogen

iii) These are specialized cells

iv) They are present in all BGA

v) They are present in certain cyanobacteria

(1) i, ii, iii, v (2) i, iii, iv

(3) i, ii, iii, iv (4) i, ii, iii, iv, v

94. A neuron at rest exhibits a stable membrane potential due to the differential distribution of ions across its axolemma. Which of the following statements accurately characterize this resting state?

A) The resting potential is primarily established by the activity of voltage – gated Na^+ and K^+ channels, which remains open.

B) The axolemma shows a high degree of impermeability to sodium ions and to the large, anionic proteins trapped within the axoplasm.

C) Relative to other ions, the membrane is significantly more permeable to potassium ions, facilitating their passive outward movement along their concentration gradient.

D) The ionic imbalance ultimately creates a positive charge of approximately + 20 mV on the inner surface of the membrane relative to the outside.

(1) A, B and D only

(2) B and C only

(3) A, B, C and D

(4) A, B and C only

95. Unique feature of Bryophytes is

(1) Dependent gametophyte

(2) Dependent sporophyte

(3) Independent gametophyte and sporophyte

(4) Independent sporophyte

96. Out of 200 – 300 million sperms in ejaculate of a human, at least how many sperms must have normal shape and size and must also show vigorous motility for normal fertility?

(1) 120 – 180 million

(2) 48 – 72 million

(3) 100 – 150 million

(4) 150 – 200 million

97. According to 5 kingdom system of classification, choose chlorophyll 'a' containing organisms that are not a part of kingdom Plantae

(1) *Spirogyra, Ulothrix, Euglena*

(2) *Cycas, Chlorella, Salvinia*

(3) *Nostoc, Anabaena, Chlorella*

(4) *Datura, Solanum, Cycas*

98. How many of the contraceptive devices mentioned below in the box do not require expert nurses/doctors for its/their insertion?

Condoms, implants, LNG – 20, lippes loop, Vaults.

Choose the correct option.

(1) Five (2) Three

(3) One (4) Two

99. Select the correct statement.

(1) Sarcomere is the anatomical unit of muscle.

(2) Contraction of a muscle fibre takes place by sliding of thick filaments over the thin filaments.

(3) Each meromyosin has a globular head with a short arm and a tail.

(4) Muscle fibres are transversely arranged in the myofibril

100. Match the column-I with column-II w.r.t aestivation

	Column-I		Column-II
A.	Lady's finger	I.	Valvate
B.	Gulmohur	II.	Vexillary
C.	Calotropis	III.	Twisted
D.	Bean	IV.	Imbricate

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-I, C-IV, D-II

(3) A-IV, B-II, C-III, D-I

(4) A-III, B-IV, C-I, D-II

101. Notochord is persistent throughout the life in
(1) Ascidia and Doliolum
(2) Petromyzon and Salpa
(3) Branchiostoma and pristis
(4) Clarias and Branchiostoma

102. Match List – I with List – II.

List – I (Type of joint)		List – II (Example)	
A.	Fibrous joint	I.	Joint between atlas and axis
B.	Cartilaginous joint	II.	Knee joint
C.	Pivot joint	III.	Sutures of skull
D.	Hinge joint	IV.	Pubic symphysis

Choose the correct answer from the options given below.
(1) A – I, B – II, C – III, D – IV
(2) A – IV, B – III, C – II, D – I
(3) A – III, B – IV, C –I, D – II
(4) A – III, B –II, C – I, D – IV

103. Palea, lemma and lodicules are present in the flowers of
(1) Fabaceae (2) Asteraceae
(3) Poaceae (4) Solanaceae

104. In which of the following organisms, any plane passing through the central axis of the body, divides the organism into two identical halves?
a) Larva of sepia
b) Adult cucumaria
c) Adult form of Ctenoplana
d) Adult form of Meandrina
e) Larva of Echinus
Select the correct option.
(1) a, b and c (2) b, c and d
(3) c, d and e (4) b, d and e

105. Among the following plants, the modified stems can act as organs of perennation in
A. *Colocasia* B. Zaminkand
C. Turmeric D. Ginger
(1) A and B only (2) C and D only
(3) A, B and C only (4) A, B, C and D

106. Given below are two statements:
Statement – I : Filariasis is not caused by any arthropod though they play an important role in its transmission.
Statement – II : Elephantiasis is a vector borne disease characterized by inflammation of lower limbs and genital organs are also often affected.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Only statement I – is correct, statement – II is incorrect
(4) Only statement I – is incorrect, statement – II is correct

107. All of the following are correct w.r.t process of ultrafiltration for urine formation, except.
(1) The glomerular capillary blood pressure causes filtration of blood through 3 cellular layers.
(2) The epithelial cells of Bowman's capsule form filtration slits.
(3) It is a non-selective process.
(4) On an average, the amount of blood filtered by the human kidneys per minute is equivalent to $1/5^{th}$ of the cardiac out put

108. Assertion (A) : Stem and leaves have conjoint type of vascular bundles.
Reason (R) : In conjoint type of vascular bundles xylem and phloem are jointly situated along same radius of vascular bundle.
(1) Both Assertion and reason are correct and reason is not correct explanation to assertion.
(2) Both Assertion and reason are correct and reason is correct explanation to assertion.
(3) Assertion is correct and reason is false
(4) Both Assertion and reason are false

109. What is commonly observed in all living prokaryotic and eukaryotic cells ?
(1) Cell wall
(2) Membrane bound nucleus
(3) Cytoplasm
(4) Membrane bound organelles

110. Choose the option which represents the steps invoved in human respiration in the correct order.
a) Utilization of oxygen by cells for catabolic reactions.
b) Transport of gases by the blood to tissues
c) Diffussion of O_2 and CO_2 between blood and tissues
d) Diffusion of O_2 and CO_2 across alveolar membrane.
(1) a, b, c, d (2) d, b, c, a
(3) b, c, a, d (4) d, a, b, c

111. Read the following and identify the characters related to wind pollinated flowers
A) Pollen grains are light and non-sticky
B) They possess well exposed stamens
C) They possess large often-feathery stigma
D) They have a single ovule in each ovary
E) Flowers are colourful and produce nectar
(1) A, B & E only
(2) B, C & D only
(3) A, B, C & E only
(4) A, B, C & D only

112. Which cell organelle is commonly seen in members of all five kingdoms of Whittaker ?
(1) Membrane bound nucleus
(2) Ribosome
(3) Plasma membrane
(4) Centrioles

113. Assertion (A) : Coacervates were probably non-cellular form of life that could have originated during evolution.
Reason (R) : Coacervates were the self duplicating aggregates of proteins surrounded by lipid layer.
(1) Both A and R are true, and R is the correct explanation of A
(2) Both A and R are true, but R is not the correct explanation of A
(3) A is true, but R is false
(4) A is false, but R is true

114. All of the following are increased in an athlete as compared to a non-athlete under resting conditions, except.
(1) Stroke volume
(2) Heart rate
(3) Duration of cardiac cycle
(4) Strength of ventricular contraction

115. Read the following statements
A) All prokaryotic cells possess cell wall surrounding plasma membrane without any exception
B) Genetic material in prokaryotic cell is naked DNA, not enveloped by a nuclear membrane
C) Specialised differentiated infolding of cell wall called mesosomes are seen in prokaryotic cell.
D) Small circular DNA outside the genomic DNA is called plasmid DNA which confers some phenotypic characters to such bacteria
Pick the correct statement
(1) B & C (2) C & D
(3) A & C (4) B & D

116. Given below are two statements:
Statement – I : All individuals have antibodies in their plasma against the antigen which is present on the surface of their RBCs.
Statement – II : All individuals have antibodies in their plasma against Rh – antigens since Birth.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Only statement I – is correct, statement – II is incorrect
(4) Only statement I – is incorrect, statement – II is correct

117. Total number of cranial nerves in frogs is equal to the total number of

(1) Digits in its hind limbs

(2) Digits in its both fore limbs and hind limbs

(3) Digits in its hind limbs multiplied by two

(4) Digits in its each fore limb multiplied by four

118. Match List-I with List-II with respect to structure of proteins.

	List-I		List-II
A.	Primary structure	I.	3-dimensional view of a protein
B.	Secondary structure	II.	An assembly of more than one polypeptide or subunits i.e., haemoglobin.
C.	Tertiary structure	III.	Linear in shape
D.	Quaternary structure	IV.	β pleated sheet

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-I, C-II, D-IV

(3) A-III, B-II, C-I, D-IV

(4) A-III, B-IV, C-I, D-II

119. Given below are two statements :

Statement-I : Primary metabolites are those having identifiable functions and play known roles in normal physiological processes.
Statement-II : Secondary metabolites are those that do not have specific identifiable role or functions in host organisms.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

120. Which of the features given below, are found in the persons suffering from down's syndrome and turner's syndrome?

i) Rudimentary ovaries

ii) Furrowed tongue

iii) Partially open mouth

iv) No menstruation

v) Gynaecomastia

vi) Small round head

	Down's syndrome	Turner's syndrome
(1)	(ii) (iii) (vi)	(i) (iv)
(2)	(ii) (iii)	(i) (iv) (vi)
(3)	(ii) (iii) (iv)	(i) (v) (vi)
(4)	(i) (iii) (iv)	(ii) (iv) (vi)

(1) 1 (2) 2

(3) 3 (4) 4

121. The relative abundance of 'A' and 'B' is higher in living organisms as compared to earth's crust. Then 'A' and 'B' respectively are

(1) A – Magnesium, B - Silicon

(2) A – Carbon, B - Hydrogen

(3) A – Silicon, B - Oxygen

(4) A – Calcium, B – Magnesium

122. Read the following statements carefully and choose the correct one with respect to population interaction?

(1) Cattle and ergets show proto-cooperation

(2) Crocodile and plover bird show commensalism.

(3) Interaction between Balanus and chathamalus is competitive co-existence

(4) Interaction between Red-billed oxpecker and black rhinoceros is proto co-operation

123. Which of the following is an incorrect match with respect to the average composition of living cell?

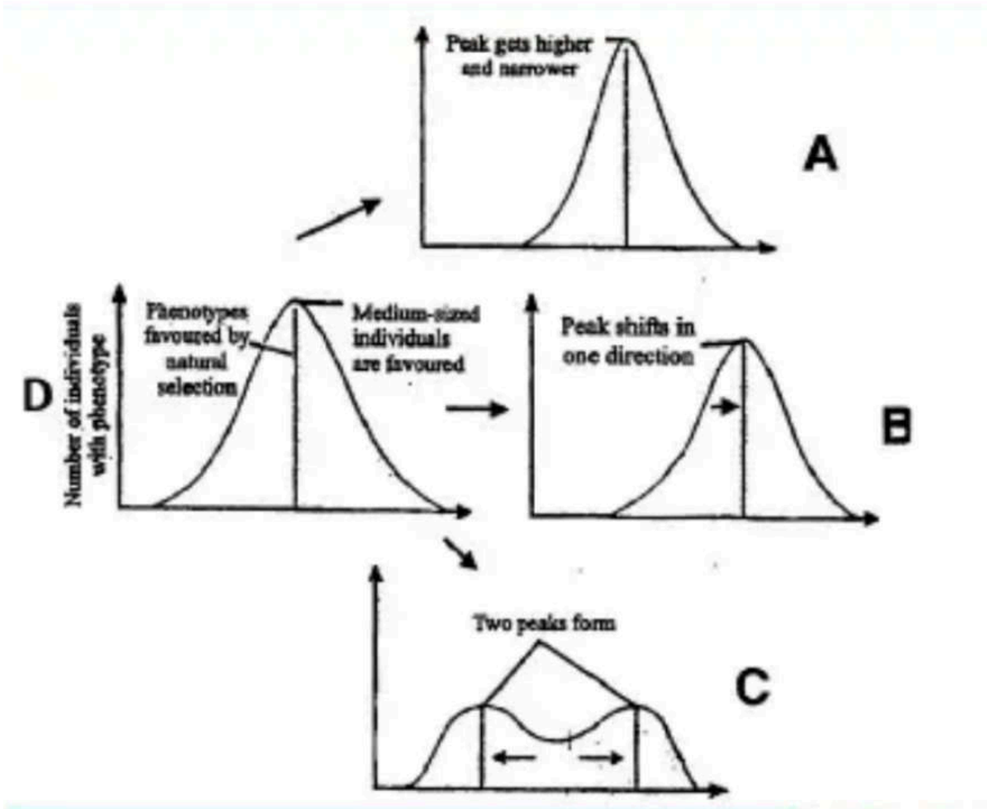
(1) Proteins: 10 – 15%

(2) Nucleic acids: 1 – 1.5%

(3) Lipids: 2%

(4) Carbohydrates: 3%

124. Identify the diagram given below and choose the correct match for it.



(1) A – Stabilizing natural selection – DDT resistance in mosquitoes

(2) B – Directional natural selection – Sickle cell anaemia

(3) C – Disruptive natural selection – Evolution of black and white snails

(4) D – Progressive natural selection – Weight of newborn babies in England

125. Select the correct statement w.r.t. periplaneta americana

(1) On an average female produce 14 – 16 ootheca, each containing 9 – 10 eggs.

(2) A pair of spermatheca is present in the 6th segment of only female cockroach

(3) Caudal styles present exclusively in females.

(4) Mesothoracic wings are opaque, dark and leathery in appearance and are useful in flight

126. Assertion (A) : Asprin reduces the risk of myo – cardial infarction and strokes.

Reason (R) : At low doses, asprin inhibits vaso-constriction and platelet aggregation.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

127. Statement-I: All cells of plant are descendents of the zygote.

Statement-II: During the cell division of a cell, DNA replication occurs but cell growth does not take place.

(1) S-I and S-II are correct

(2) S-I and S-II are incorrect

(3) S-I is correct, S-II is incorrect

(4) S-I is incorrect, S-II is correct

128. During cell cycle, M phase starts with ___ usually ends with ____respectively

(1) synapsis and crossing over

(2) karyokinesis and telophase

(3) pachytene tetrad and segregation of homologous chromosomes

(4) karyokinesis and cytokinesis

129. Given below are five matches of animals with respiratory organs.

A – Pterophyllum – Gills

B – Hyla – skin

C – Chelone – Lungs

D – Hippocampus – Trachea

E – Ascidia – Book lungs

The correct matches are

(1) A and D

(2) A, B, C and D

(3) A, B and C

(4) B, C and E only

130. Two kinetochores of a centromere of a chromosome with two sister chromatids connected to spindle fibers coming from opposite poles of a cell is seen during
(1) Metaphase-I and Metaphase-II
(2) Metaphase of mitosis and Metaphase-I
(3) Metaphase of Mitosis and Anaphase-I
(4) Metaphase of Mitosis and Metaphase-II

131. In a population Allele T, represents the ability to taste a particular chemical is dominant over allele t, for the inability to taste the chemical. Four hundred university students were surveyed and 64 students were found to be non – tasters. Calculate the percentage of heterozygous students. Assume that the population in Hardy – Weinberg equilibrium.
(1) 40 % (2) 36 %
(3) 60 % (4) 48 %

132. During mitosis in an animal cell, each aster is with
A) a centrosome
B) microtubules radiating out
C) two spindle apparatii
D) two centrioles
The correct is
(1) A,B and C (2) A,B and D
(3) B,C and D (4) C and D only

133. Which of following animal contains "Radula" in mouth?
(1) Sea – urchin (2) Sea – mouse
(3) Sea – fan (4) Sea – hare

134. Select the incorrect match w.r.t hormone, its source gland and function.

	Hormone	Source	Function
(1)	Melatonin	Pineal gland	Regulates body temperature
(2)	ADH	Hypo thalamus	Conservation of water
(3)	Androgen	Testis	Erythropioesis
(4)	CCK	Gastro – intestinal tract	Stimulates secretion of water and bicarbonate ions from pancreas

135. During C_4 cycle, malic acid or aspartic acid is formed in
(1) cytoplasm of mesophyll cells
(2) chloroplast of mesophyll cells
(3) cytoplasm of bundle sheath cells
(4) chloroplast of bundle sheath cells

136. Which of the following options correctly represent the relationship between germ layers and body cavity in different animal groups?
(1) All acoelomates are triploblastic
(2) All pseudo coelomates are diploblastic
(3) All coelomates are diploblastic.
(4) Only triploblastic animals can be coelomate (or) pseudo – coelomate

137. During Z scheme, ___ are transported across the membrane as electrons move through the photosystems
(1) Water (2) Oxygen
(3) Electrons (4) Protons

138. Protons accumulated in the lumen of thylakoids during light reaction are taken from
I) stroma II) water
III) Oxygen IV) Carbondioxide
(1) I and IV (2) I and II
(3) I,II and III (4) II,III and IV

139. Match List – I with List – II and choose the correct option w.r.t embryonic development.

List – I		List – II	
A.	Eyelashes formed	I.	End of 8th week
B.	Limbs and digits formed	II.	End of 12th week
C.	External genital organs	III.	End of 24 weeks
D.	First movements of the foetus	IV.	End of 5th month

Choose the correct answer from the options given below.

(1) A – I, B – II, C – IV, D – III

(2) A – II, B – I, C – III, D – IV

(3) A – III, B – I, C –II, D – IV

(4) A – IV, B –III, C – II, D – I

140. Assertion : Cytochrome c of ETS of respiration is a mobile electron carrier.

Reason : Cytochrome c picks up electrons from complex III and conveys them directly to oxygen.

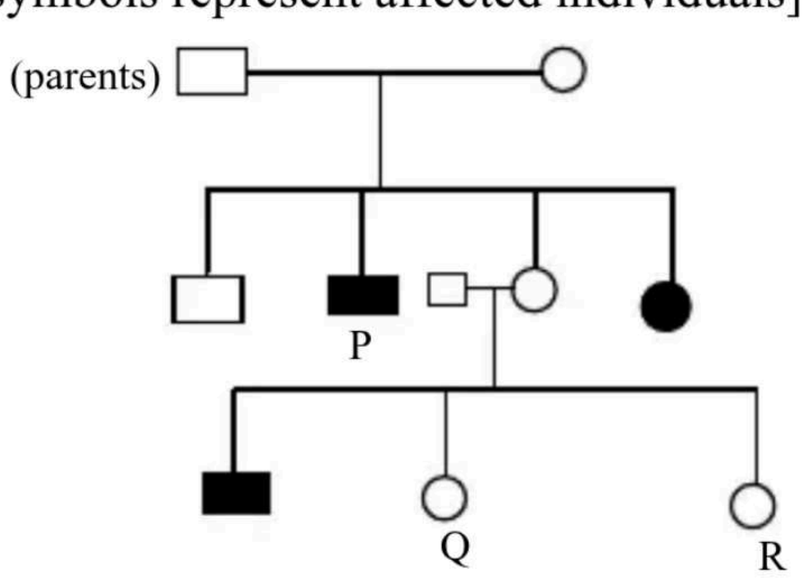
(1) Both assertion and reason are correct and reason correctly explains the assertion.

(2) Both assertion and reason are correct and reason does not correctly explain the assertion

(3) Assertion is correct and reason is incorrect.

(4) Both assertion and reason are incorrect.

141. Study the given pedigree chart and choose the correct statement based on it. [The shaded symbols represent affected individuals]



(1) The trait under the study can be haemophilia

(2) Both parents are heterozygous for the trait

(3) The genotype of (P) can be Aa and shows myotonic dystrophy.

(4) Females (Q) and (R) are not affected because parents are not carrying the allele of the trait

142. The source of phosphate for PGAL phosphorylation during glycolysis is

(1) ATP (2) PGA

(3) NADPH (4) H_3PO_4

143. Given below are two statements:

Statement – I : The solid embryo with blastomeres is called morula.

Statement – II : The size of morula in human is smaller than that of the fertilized egg.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Only statement I – is correct, statement – II is incorrect

(4) Only statement I – is incorrect, statement – II is correct

144. Which of the following regulates the kidney functions by increasing the glomerular filtration rate?

a) Aldosterone

b) Anti – Diuretic hormone

c) Atrial natri uretic factor

d) Angiotensin – II

(1) b and c only

(2) a and b only

(3) a, b and d only

(4) Only 'a'

145. Which one of the PGR would you use if you are asked to induce immediate stomatal closure in leaves ?

(1) Auxin (2) ABA

(3) GA (4) Cytokinin

146. Match List – I with List – II

List – I		List – II	
A.	Innate immunity	I.	Immuno – globulins
B.	Cell mediated immunity	II.	Interferons
C.	Antibody mediated immunity	III.	PMNL
D.	Neutrophils	IV.	Cytotoxic T - cells

Choose the correct answer from the options given below.

(1) A – I, B – II, C – III, D – IV

(2) A – II, B – I, C – IV, D – III

(3) A – III, B – II, C –IV, D – I

(4) A – II, B –IV, C – I, D – III

147. Plant growth regulators are

(1) intercellular intrinsic factors

(2) intracellular intrinsic factors

(3) intercellular extrinsic factors

(4) intracellular extrinsic factors

148. Select the incorrect statement regarding Variable Number of Tandem Repeats (VNTR)

(1) VNTR belongs to a class of satellite DNA referred to as mini – satellite.

(2) The number of repeats varies from chromosome to chromosome in an individual

(3) They show very high degree of polymorphism.

(4) Number of VNTR same in all individuals of population

149. Match Column-I with Column-II and select the correct option from the codes given below.

Column-I	Column-II
A. Auxins	(i) Breaking seed dormancy
B. Gibberellins	(ii) Inducing fruit ripening
C. Cytokinins	(iii) Formation of abscission layer
D. Ethylene	(iv) Root initiation
	(v) Chloroplast development in leaves

(1) A-(iv), B-(i), C-(v), D-(ii)

(2) A-(iv), B-(v), C-(iii), D-(ii)

(3) A-(i), B-(iii), C-(ii), D-(iv)

(4) A-(iii), B-(iv), C-(i), D-(v)

150. Activated sludge is

(1) Settled anaerobic bacterial flocs after biological treatment

(2) Released into running water bodies

(3) Formed after activity of anaerobic microbes in anaerobic sludge digester

(4) Settled mass of aerobic bacteria and fungi

151. In early 20th – century evolutionary biology, hugo devries, based on his work on evening primrose, proposed that mutations are sudden, large, random and directional, and responsible for speciation. He believed that evolution occurs through mutations, not through the gradual, small, heritable variations emphasized by Darwin.

Choose the correct answer from the options given below.

(1) The statement is correct with respect to Hugo devries theory but incorrect for Darwin's conept of evolution.

(2) The statement is incorrect for both Hugodevries and Darwin concept of evolution

(3) The statement is correct for both Hugo devries and Darwin's concept of evolution

(4) The statement is correct with respect to Darwin but incorrect for Hugo devries concept of evolution

152. Assertion (A) : A population has certain attributes that an individual does not.

Reason (R) : An individual may have birth rates and death rates, but a population has birth and death.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

153. How many of the following statements are incorrect regarding uses of GM crops

I. increased reliance on chemical pesticides

II. helped to reduce post harvest losses

III. their cultivation leads to early exhaustion of fertile soil

IV. enhanced nutritional value of food

V. made crops more tolerant to abiotic stress

(1) 3 (2) 4

(3) 2 (4) 5

154. Match the organism with its use in biotechnology.

	Organism		Use in biotechnological procedures
A.	*Bacilllus thuringiensis*	I.	Cloning vector
B.	*Thermus aquaticus*	II.	Construction of first rDNA molecule
C.	*Agrobacterium tumefaciens*	III.	Taq polymerase
D.	*Salmonella typhimurium*	IV.	Cry proteins

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-II, C-IV, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-IV, C-III, D-I

155. Choose the incorrect match w.r.t humans.

(1) MALT – Consitutes about 50% of the lymphoid tissue in our body.

(2) Peyer's patches – Primary lymphoid organs that are present in our small intestine

(3) Lymph nodes – Small solid structures located at different points along the lymphatic system.

(4) Thymus – Lobed organ located near the heart and beneath the breast bone

156. Which of the following events took place before 1990?

(1) An American company prepared genetically engineered human insulin.

(2) First clinical gene therepy was given to a 4 year old girl with a genetic disorder.

(3) First transgenic cow was made that produced human protein – enriched milk.

(4) An American company got patent rights over Basmati rice

157. Statement-I: In order to link the alien DNA, the vector needs to have very few, preferably single recognition site for each type of restriction enzyme.

Statement-II: Presence of more than one type of recognition sequences within vector will generate several fragments which will complicate the gene cloning

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

158. Match List – I with List – II

List – I		List – II	
A.	Sacred groves	I.	Alien species
B.	Cryopreservation	II.	High degree of endemism
C.	Eicchornia	III.	Ex-situ conservation
D.	Bio-diversity hot spots	IV.	Khasi hills in meghalaya

Choose the correct answer from the options given below.

(1) A – IV, B – III, C –I, D – II

(2) A – II, B – IV, C – I, D – III

(3) A – IV, B – I, C –II, D – III

(4) A – IV, B –III, C – II, D – I

159. The restriction enzyme employed to deliver a desired gene at tet^R site of pBR322 is

(1) *Bam* HI (2) *Sal* I

(3) *Pvu* II (4) 1 (or) 2

160. Which of the following statements are correct for cancer cells?

a) Do not show the property of contact inhibition.

b) Cell growth and differentiation is highly controlled

c) Actively divide and compete with normal cells for vital nutrients.

d) Remain confined to their original location

e) Some of the tumor cells break off and reach distant sites and start forming new tumor there.

(1) Only a and e (2) Only b, d, e

(3) Only a, c, e (4) All are correct

161. The term that describes the species that are at risk of extinction in the wild in the near future is

(1) Extinct species

(2) Endangered species

(3) Vulnerable species

(4) Endemic species

162. Given below are two statements:

Statement – I : Natural killer cells destroy infected body cells and cancer cells by phagocytosis.

Statement – II : B – lymphocytes and T – lymphocytes are useless for a person unless and until they have antigen specific receptors on them.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Only statement I – is correct, statement – II is incorrect

(4) Only statement I – is incorrect, statement – II is correct

163. The source of Taq polymerase used in PCR technique is a

(1) Thermophilic blue green algae

(2) A Protistan

(3) A Fungus

(4) Thermophilic bacteria (Eubacteria)

164. The host is cultured in a continuous culture system, where used medium is drained out from one side while fresh medium is added from other side to maintain cells in their physiologically _____ phase

(1) Stationary phase (2) Log phase

(3) Asymptote (4) Dormant

165. In eukaryotes chromosomes are composed of

A. DNA

B. RNA

C. Histone

D. Non-histone

E. Nucleosome

(1) A, B only

(2) C, D, E only

(3) A, C, E only

(4) A, B, C, D, E

166. In an Ecosystem, the amount of energy utilized by plants for respiration is 150 kJ, then find out N.S.P in secondary carnivores [Note – energy utilized for respiration – 25%]

(1) 4.5 k.J (2) 0.045 k.J

(3) 45 k.J (4) 0.45 k.J

167. Select the answer which correctly matches the endo-crine gland with the hormones it secretes and disease due to its hypo secretion.

	Endocrine gland	Hormone	Disease due to hypo-secretion
(1)	Pituitary gland	Growth hormone	Acromegaly
(2)	Ovary	Estrogen	Gynaecomastia
(3)	Aderenal gland	Oxytocin	Addison's disease
(4)	Thyroid gland	Thyroxine	Myxoedema

168. The medical termination of pregnancy (Amendment) Act, 2017 was enacted by the Government of India with all of the following intensions, except.

(1) Reducing the incidence of illegal abortion.

(2) Reducing maternal mortality

(3) Reducing maternal morbidity

(4) Increasing the incidence of female foeticide

169. In a sample of DNA, the base composition (mole ratio) is found to be A = 40, T = 22, G = 21 and C=17 . This suggests that

(1) DNA is a linear duplex

(2) DNA is a circular duplex

(3) DNA is single-stranded

(4) DNA has a high melting point

170. Assertion (A) : In a food chain members of successive higher levels are fewer in number.

Reason (R) : Number of organisms at any trophic level depends upon the availability of organisms which serve as food at the lower level.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

171. Incorrect match among the following with respect to protein synthesis in bacteria is

(1) mRNA-Provides the template

(2) tRNA-brings amino acid and reads the genetic code

(3) rRNA-plays structural and catalytic role

(4) It doesn't occur in any cell organelle

172. When a colourblind woman who is also suffering from sickle cell anaemia married a non-colourblind man who is also suffering from sickle cell anaemia. What is the probability of occurrence of both these traits in their progeny?

(1) 50% of daughters are colourblind and also suffering from sickle cell anaemia.

(2) 50% of daughters and 50% of sons are colourblind and also suffering from sickle cell anaemia.

(3) All sons and daughers are normal, for colourblindness but suffering from sickle cell anaemia.

(4) All sons are colourblind and also suffering from sickle cell anaemia

173. Read the following statements :
Statement-I : The split-gene arrangement probably represents an ancient feature of the genome.
Statement-II : The presence of introns is reminiscent of antiquity and the process of splicing represents the dominance of the RNA-world.
In light of the given statements, choose the correct answer from the options given below.
(1) Both S-I and S-II are correct.
(2) Both S-I and S-II are incorrect.
(3) S-I is correct but S-II is incorrect.
(4) S-I is incorrect but S-II is correct.

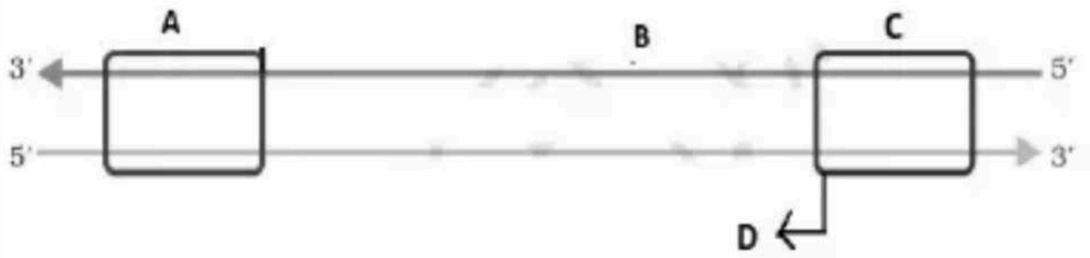
174. Assertion (A) : Periodic infusion of genetically engineered insulin is required to treat the hereditary disease caused by the deficiency of ADA.
Reason (R) : The first clinical gene therapy was given to a 4 year old girl suffering with ADA deficiency.
(1) Both A and R are true, and R is the correct explanation of A
(2) Both A and R are true, but R is not the correct explanation of A
(3) A is true, but R is false
(4) A is false, but R is true

175. Assertion (A) : Insertion (or) deletion of one (or) two bases, changes the reading frame from that point of insertion or deletion
Reason (R) : The disease sickle cell anaemia is caused due to point mutation.
(1) Both A and R are correct. R is the correct explanation of A.
(2) Both A and R are correct. R is not the correct explanation of A.
(3) A is true, but R is false
(4) A is false, but R is true

176. If the ratio of (T+C)/(A+G) in one strand of DNA is 1.43, same ratio in the complementary strand is :
(1) 0.35 (2) 0.70
(3) 1.43 (4) 2.10

177. Mark the correct sequence of events that take place during the formation of embryo sac.
(1) Megaspore→ megaspore mother cell→ embryo sac
(2) Nucellus→ Archesporium → megaspore mother cell→ megaspore→ embryo sac
(3) Megaspore mother cell→ megasporangium→megaspores→ egg
(4) Megasporophyll→ ovule→ ovary→ embryo sac.

178. Study the following diagram and identify A, C, D, B respectively.



(1) transcription start site, RNA polymerase binding site, Transcription end site, coding strand
(2) Transcription end site, coding strand, transcription start site, RNA polymerase binding site
(3) Transcription end site, RNA polymerase binding site, transcription start site, coding strand
(4) RNA polymerase binding site, Transcription end site, transcription start site, coding strand

179. A heterogenous pink flowered *Antirrhinum* plant on self-pollination also produced 125 white flowered plants. What is the number of plants with pink flowers ?
(1) 125 (2) 250
(3) 375 (4) 400

180. A man with B blood group marries a woman with an AB blood group. Which blood group will never appear among the progeny ?
(1) A (2) B
(3) AB (4) O